Given (for all derivations below)

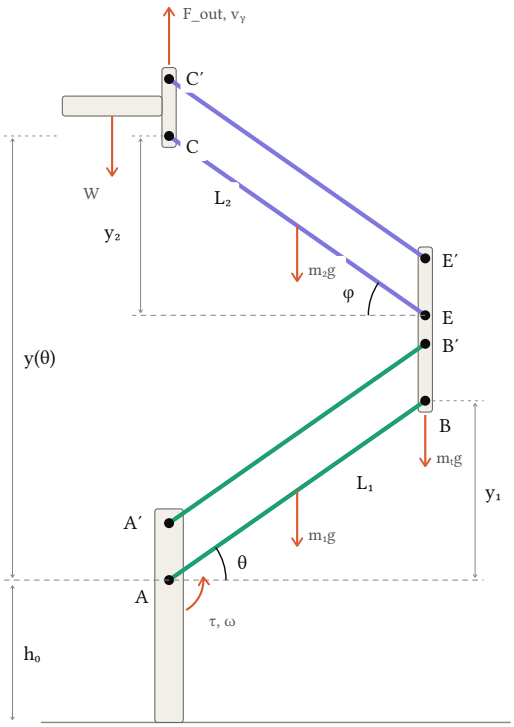
$\varphi = \theta$

$L = L_1 = L_2$

$\omega := \dot{\theta}$ and $\alpha := \dot{\omega} = \ddot{\theta}$

$v := \dot{y}$

$\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$



Goal 1

Find a function v, linear velocity, for the top of the lift

1. $y(\theta) = y_1 + BB' + B'E + y_2$ by the segment addition postulate
2. $\sin\theta = \frac{y_1(\theta)}{L_1}$ and $\sin\varphi = \frac{y_2(\varphi)}{L_2}$ by the definition of sin in right angle trigonometry
3. $y_1(\theta) = L_1 \sin\theta$ and $y_2(\varphi) = L_2 \sin\varphi$ by multiplication
4. $y(\theta) = L_1 \sin\theta + BB' + B'E + L_2 \sin\varphi$ by substitution
5. $y(\theta) = L \sin\theta + BB' + B'E + L \sin\theta$ by substitution
6. $y(\theta) = 2L \sin\theta + BB' + B'E$ by substitution
7. $v(\theta, \omega) = 2L\omega \cos\theta$ by differentiation

Note: It can be shown trivially that the horizontal component of velocity would be 0 as $\theta = \varphi$ and $L_1 = L_2$

Goal 2

Given

The origin is at A, with $+x$ to the right and $+y$ up.

$m_1 :=$ total mass of the two lower bars

$m_2 :=$ total mass of the two upper bars

$m_t :=$ mass of the middle tower

$m_{\text{car}} :=$ mass of the carriage (top tower from C to C', intake, and payload)

$b := BB' + B'E$, the fixed vertical distance from B to E

$h_t :=$ fixed height of the middle tower's center of mass above B

$h_{\text{car}} :=$ fixed height of the carriage's center of mass above C

$g :=$ gravitational acceleration

For any point P, $\vec{r}_P :=$ the position vector of P relative to A

For any point P, $\vec{v}_P := \dot{\vec{r}}_P$

$\vec{r}_1$, $\vec{r}_t$, $\vec{r}_2$, and $\vec{r}_{\text{car}}$: positions of the centers of mass of the lower bars, middle tower, upper bars, and carriage.

Find a function $\tau(\theta, \omega, a)$, the torque required at A for the top of the lift to have vertical acceleration a, given angle θ and angular velocity ω.

1. $\vec{r}_B = (L\cos\theta, L\sin\theta)$ by the definitions of sin and cos in right angle trigonometry
2. $\vec{r}_E = \vec{r}_B + (0, b)$ by vector addition
3. $\vec{r}_E = (L\cos\theta, L\sin\theta + b)$ by substitution
4. $\overrightarrow{EC} = (-L\cos\varphi, L\sin\varphi)$ by the definitions of sin and cos in right angle trigonometry
5. $\overrightarrow{EC} = (-L\cos\theta, L\sin\theta)$ by substitution
6. $\vec{r}_C = \vec{r}_E + \overrightarrow{EC}$ by vector addition
7. $\vec{r}_C = (0, 2L\sin\theta + b)$ by substitution of (3) and (5) into (6)
8. $\vec{r}_1 = \frac{1}{2}\vec{r}_B = \left(\frac{L}{2}\cos\theta, \frac{L}{2}\sin\theta\right)$ by the fact that the center of mass of a uniform bar is its midpoint

   Note: bar $A'B'$ is bar AB translated by the constant vector $\overrightarrow{AA'}$ (parallelogram), so its center is $\vec{r}_1$ plus a constant. Both lower bars $\therefore$ have the same velocity and angular velocity and are treated as one body of mass m_1. The same argument applies to the upper bars.
9. $\vec{r}_t = \vec{r}_B + (0, h_t) = (L\cos\theta, L\sin\theta + h_t)$ by vector addition and the definition of h_t
10. $\vec{r}_2 = \vec{r}_E + \frac{1}{2}\overrightarrow{EC}$ by the fact that the center of mass of a uniform bar is its midpoint
11. $\vec{r}_2 = \left(\frac{L}{2}\cos\theta, \frac{3L}{2}\sin\theta + b\right)$ by substitution of (3) and (5)
12. $\vec{r}_{\text{car}} = \vec{r}_C + (0, h_{\text{car}}) = (0, 2L\sin\theta + b + h_{\text{car}})$ by vector addition, the definition of h_{car}, and (7)
13. $\vec{v}_1 = \left(-\frac{L}{2}\omega\sin\theta, \frac{L}{2}\omega\cos\theta\right)$ by differentiation of (8)
14. $|\vec{v}_1|^2 = \frac{L^2\omega^2}{4}\left(\sin^2\theta + \cos^2\theta\right)$ by the definition of magnitude
15. $|\vec{v}_1|^2 = \frac{L^2\omega^2}{4}$ by substitution

16. $\vec{v}_t = (-L\omega\sin\theta, L\omega\cos\theta)$ by differentiation of (9)
17. $|\vec{v}_t|^2 = L^2\omega^2(\sin^2\theta + \cos^2\theta)$ by the definition of magnitude
18. $|\vec{v}_t|^2 = L^2\omega^2$ by substitution
19. $\vec{v}_2 = \left(-\frac{L}{2}\omega\sin\theta, \frac{3L}{2}\omega\cos\theta\right)$ by differentiation of (11)
20. $|\vec{v}_2|^2 = \frac{L^2\omega^2}{4}(\sin^2\theta + 9\cos^2\theta)$ by the definition of magnitude
21. $|\vec{v}_2|^2 = \frac{L^2\omega^2}{4}(1 + 8\cos^2\theta)$ by substitution
22. $\vec{v}_{\text{car}} = (0, 2L\omega\cos\theta)$ by differentiation of (12)
23. $|\vec{v}_{\text{car}}|^2 = 4L^2\omega^2\cos^2\theta$ by the definition of magnitude

For a rigid body moving in a plane, its kinetic energy is $T := \frac{1}{2}m|\vec{v}_{\text{cm}}|^2 + \frac{1}{2}I_{\text{cm}}\omega^2$, where I_{cm} is its moment of inertia about its center of mass. For a uniform bar of mass m and length L, $I_{\text{cm}} = \frac{1}{12}mL^2$. The two bars in each pair have the same velocity and angular velocity, so their kinetic energies add to the formula with their total mass.

Note: All four bars have angular velocity $\pm\omega$, so the same ω^2. The middle tower and carriage translate without rotating, so they have no rotational kinetic energy.

24. $I_1 = \frac{1}{12}m_1L^2$ by the moment of inertia of a uniform bar
25. $I_2 = \frac{1}{12}m_2L^2$ by the moment of inertia of a uniform bar
26. $T_t = \frac{1}{2}m_t|\vec{v}_t|^2$ and $T_{\text{car}} = \frac{1}{2}m_{\text{car}}|\vec{v}_{\text{car}}|^2$ since the tower and carriage do not rotate
27. $T_1 = \frac{1}{2}m_1\frac{L^2\omega^2}{4} + \frac{1}{2}\left(\frac{1}{12}m_1L^2\right)\omega^2$ by substitution of (15) and (24)
28. $T_1 = \frac{1}{2}\left(\frac{m_1}{3}\right)L^2\omega^2$ by substitution
29. $T_t = \frac{1}{2}m_tL^2\omega^2$ by substitution of (18) into (26)
30. $T_2 = \frac{1}{2}m_2\frac{L^2\omega^2}{4}(1 + 8\cos^2\theta) + \frac{1}{2}\left(\frac{1}{12}m_2L^2\right)\omega^2$ by substitution of (21) and (25)
31. $T_2 = \frac{1}{2}m_2\left(\frac{1}{3} + 2\cos^2\theta\right)L^2\omega^2$ by substitution
32. $T_{\text{car}} = \frac{1}{2}m_{\text{car}}(4\cos^2\theta)L^2\omega^2$ by substitution of (23) into (26)
33. $T = T_1 + T_t + T_2 + T_{\text{car}}$ by the definition of total kinetic energy
34. $T = \frac{1}{2}L^2\left[\frac{m_1+m_2}{3} + m_t + 2(m_2 + 2m_{\text{car}})\cos^2\theta\right]\omega^2$ by substitution of (28), (29), (31), (32)
35. $M := \frac{m_1+m_2}{3} + m_t$
36. $I_{\text{eff}}(\theta) := L^2[M + 2(m_2 + 2m_{\text{car}})\cos^2\theta]$
37. $T = \frac{1}{2}I_{\text{eff}}(\theta)\omega^2$ by substitution of (35) and (36) into (34)
38. $I'_{\text{eff}}(\theta) = -4(m_2 + 2m_{\text{car}})L^2\sin\theta\cos\theta$ by differentiation of (36), since $\frac{\mathrm{d}}{\mathrm{d}\theta}\cos^2\theta = -2\sin\theta\cos\theta$

Let $V = g\Big(m_1[\vec{r}_1]_y + m_t[\vec{r}_t]_y + m_2[\vec{r}_2]_y + m_{\text{car}}[\vec{r}_{\text{car}}]_y\Big)$ be the gravitational potential energy, where $[\vec{r}_P]_y$ is the y-component of $\vec{r}_P$. (The second bar in each pair only adds a constant, which does not affect $V'(\theta)$.)

39. $V = g\left[m_1\frac{L}{2}\sin\theta + m_t(L\sin\theta + h_t) + m_2\left(\frac{3L}{2}\sin\theta + b\right) + m_{\text{car}}(2L\sin\theta + b + h_{\text{car}})\right]$ by substitution of (8), (9), (11), (12)
40. $V = g\left[m_1\frac{L}{2}\sin\theta + m_tL\sin\theta + m_th_t + \frac{3m_2L}{2}\sin\theta + m_2b + 2m_{\text{car}}L\sin\theta + m_{\text{car}}b + m_{\text{car}}h_{\text{car}}\right]$ by substitution

41. $V = g\Big[\Big(m_1\frac{L}{2}\sin\theta + m_t L\sin\theta + \frac{3m_2 L}{2}\sin\theta + 2m_{\text{car}}L\sin\theta\Big) + (m_t h_t + m_2 b + m_{\text{car}}b + m_{\text{car}}h_{\text{car}})\Big]$ by substitution
42. $V = g\Big[L\sin\theta\Big(\frac{m_1}{2} + m_t + \frac{3m_2}{2} + 2m_{\text{car}}\Big) + (m_t h_t + m_2 b + m_{\text{car}}(b + h_{\text{car}}))\Big]$ by substitution
43. $V = gL\sin\theta\Big(\frac{m_1}{2} + m_t + \frac{3m_2}{2} + 2m_{\text{car}}\Big) + g(m_t h_t + m_2 b + m_{\text{car}}(b + h_{\text{car}}))$ by substitution
44. $V_0 \coloneqq g(m_t h_t + m_2 b + m_{\text{car}}(b + h_{\text{car}}))$
45. $V = gL\sin\theta\Big(\frac{m_1}{2} + m_t + \frac{3m_2}{2} + 2m_{\text{car}}\Big) + V_0$ by substitution of (44) into (43)
46. $K \coloneqq \frac{m_1}{2} + m_t + \frac{3m_2}{2} + 2m_{\text{car}}$
47. $V'(\theta) = gKL\cos\theta$ by differentiation of (45), using (46)
48. $\tau\omega = \dot{T} + \dot{V}$ by the work-energy theorem
49. $\dot{T} = \frac{1}{2}I'_{\text{eff}}(\theta)\omega^3 + I_{\text{eff}}(\theta)\omega\alpha$ by differentiation of (37) using the product and chain rules
50. $\dot{T} = \omega\big(I_{\text{eff}}\alpha + \frac{1}{2}I'_{\text{eff}}\omega^2\big)$ by substitution
51. $\dot{V} = V'(\theta)\omega$ by the chain rule
52. $\tau\omega = \omega\big(I_{\text{eff}}\alpha + \frac{1}{2}I'_{\text{eff}}\omega^2 + V'(\theta)\big)$ by substitution of (50) and (51) into (48)
53. $\tau = I_{\text{eff}}\alpha + \frac{1}{2}I'_{\text{eff}}\omega^2 + V'(\theta)$ by division by ω

    Note: this requires $\omega \neq 0$. Both sides are continuous in time, so (53) also holds at isolated instants where $\omega = 0$, such as starting from rest.
54. $v = 2L\omega\cos\theta$ by Goal 1, step 7
55. $a = 2L(\alpha\cos\theta - \omega^2\sin\theta)$ by differentiation of (54)
56. $\alpha = \frac{a + 2L\omega^2\sin\theta}{2L\cos\theta}$ by solving (55) for α, which is valid since $\cos\theta \neq 0$ and $L \neq 0$
57. $\tau = L^2\big[M + 2(m_2 + 2m_{\text{car}})\cos^2\theta\big]\frac{a + 2L\omega^2\sin\theta}{2L\cos\theta} - 2(m_2 + 2m_{\text{car}})L^2\omega^2\sin\theta\cos\theta + gKL\cos\theta$ by substitution of (36), (38), (47), (56) into (53)
58. $L^2\big[M + 2(m_2 + 2m_{\text{car}})\cos^2\theta\big]\frac{a + 2L\omega^2\sin\theta}{2L\cos\theta} = L\big[M + 2(m_2 + 2m_{\text{car}})\cos^2\theta\big]\frac{a + 2L\omega^2\sin\theta}{2\cos\theta}$ by substitution
59. $L\big[M + 2(m_2 + 2m_{\text{car}})\cos^2\theta\big]\frac{a + 2L\omega^2\sin\theta}{2\cos\theta} = L\big(\frac{M}{2\cos\theta} + (m_2 + 2m_{\text{car}})\cos\theta\big)(a + 2L\omega^2\sin\theta)$ by substitution
60. $L\big(\frac{M}{2\cos\theta} + (m_2 + 2m_{\text{car}})\cos\theta\big)(a + 2L\omega^2\sin\theta) = L\big(\frac{M}{2\cos\theta} + (m_2 + 2m_{\text{car}})\cos\theta\big)a + 2L^2\omega^2\sin\theta\big(\frac{M}{2\cos\theta} + (m_2 + 2m_{\text{car}})\cos\theta\big)$ by substitution
61. $2L^2\omega^2\sin\theta\big(\frac{M}{2\cos\theta} + (m_2 + 2m_{\text{car}})\cos\theta\big) = L^2\omega^2(M\tan\theta + 2(m_2 + 2m_{\text{car}})\sin\theta\cos\theta)$ by substitution
62. $\tau = L\big(\frac{M}{2\cos\theta} + (m_2 + 2m_{\text{car}})\cos\theta\big)a + ML^2\omega^2\tan\theta + 2(m_2 + 2m_{\text{car}})L^2\omega^2\sin\theta\cos\theta - 2(m_2 + 2m_{\text{car}})L^2\omega^2\sin\theta\cos\theta + gKL\cos\theta$ by substitution of (58)–(61) into (57)
63. $\tau = L\big(\frac{M}{2\cos\theta} + (m_2 + 2m_{\text{car}})\cos\theta\big)a + ML^2\omega^2\tan\theta + gKL\cos\theta$ by substitution
64. $\boxed{\tau(\theta, \omega, a) = L\big[\big(\frac{M}{2\cos\theta} + (m_2 + 2m_{\text{car}})\cos\theta\big)a + ML\omega^2\tan\theta + gK\cos\theta\big]}$ by substitution

Goal 3

Find $\alpha(\theta, \omega)$, the angular acceleration of the lower stage given angle θ and angular velocity ω

Goal 4

Find the time to raise the lift from an angle θ_0 to θ_f